

# FORMALLY VERIFIED CONGRUENCES FOR APÉRY-LIKE SEQUENCES, AND AN ARITHMETIC NOTE ON THE BROWN–ZUDILIN DENOMINATORS

**ABSTRACT.** We report a Lean 4 development, checked by the Lean kernel and free of `sorry`, that proves: (i) the Lucas congruence  $a_{ap+r} \equiv a_a a_r \pmod{p}$  for Apéry’s  $\zeta(3)$  numbers, for *every* prime and every  $a \geq 0$ , together with its base- $p$  digit-product form, and the same pair of statements for the Brown–Zudilin double sum  $Q_n = \sum_{k,l} T(n, k, l)$ ; (ii) a descent lemma for truncated twisted zeta values,  $p^r K_\chi^{(r)}(y) \equiv \chi(p) K_\chi^{(r)}(\lfloor y/p \rfloor)$ , and its extension to monomials; (iii) an abstract congruence for the *second* solution of an Apéry-like recurrence,  $p^w B(ap+r) \equiv \chi_p B(a) A(r) \pmod{p}$ , under seven explicit hypotheses on the summand and on a harmonic weight of uniform degree  $w$ , with the character value  $\chi_p$  carried as a parameter; (iv) two instances of it, at  $w = 3$  for Apéry’s  $\zeta(3)$  pair and at  $w = 2$  for Franel’s, both instantiated at  $\chi_p = 1$ ; (v) the closed form  $\sum_{k \leq n} \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}) = b_n$ , where  $b_n$  is the sequence *defined* by Apéry’s recurrence with  $b_0 = 0$ ,  $b_1 = 6$ , and, as a consequence, the second-row congruence  $p^3 b_{ap+r} \equiv b_a a_r \pmod{p}$  for that classical companion sequence itself, for every prime  $p$  and all  $a, r < p$ ; and (vi) a small arithmetic fact taking Brown–Zudilin’s printed value  $P_2 = 1190161/384$  as a quoted hypothesis:  $d_2^5 P_2 \notin \mathbb{Z}$ ,  $12 d_2^5 P_2 \in \mathbb{Z}$ , and — minimality — any  $c \in \mathbb{N}$  with  $c d_2^5 P_2 \in \mathbb{Z}$  is divisible by 12. Each statement below is accompanied by the name of the declaration that proves it, so that it can be checked against the repository; the paper claims nothing that the kernel has not checked. In particular nothing about the *value*  $\zeta(3)$  is formalised anywhere: this is a paper about congruences satisfied by certain sequences.

## 1. INTRODUCTION AND SCOPE

Let  $a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$  be Apéry’s numbers for  $\zeta(3)$  (OEIS A005259). They satisfy a congruence of Lucas type,

$$a_{ap+r} \equiv a_a a_r \pmod{p} \quad (p \text{ prime}, a \geq 0, 0 \leq r < p), \quad (1.1)$$

hence  $a_n \equiv \prod_i a_{n_i} \pmod{p}$  over base- $p$  digits. This is classical [3], and Malik and Straub [4] prove it for all fifteen sporadic Apéry-like sequences, Straub [9] sharpening it to modulus  $p^2$ . Apéry’s irrationality proof, however, uses a *second* solution  $b_n$  of the same recurrence, and for that row no congruence of Lucas type appears in the literature.

This paper reports what has been proved about the second row *and machine-checked*. The scope rule is strict and we state it once, because it is the point of the paper:

Every mathematical assertion in Sections 2–6 is the informal reading of a Lean 4 declaration that the Lean kernel has checked, with no `sorry`, no `admit`, no `native_decide`, and no axioms beyond `propext`, `Classical.choice` and `Quot.sound`. Each is displayed with the name of its declaration. Nothing that has been checked only numerically, or proved only on paper, is stated here as a result.

Table 1 is the inventory. §7 says how to rebuild and re-check the development in full.

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TABLE 1. The development. Each module is the file `ZetaLucas/⟨module⟩.lean`, and every declaration listed lives in the `ZetaLucas` namespace with the single exception of `Finset.sum_range_mul`.

| module       | principal declarations                                                                                                                                                                                     | § |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Core         | <code>choose_digits</code> , <code>choose_digits'</code> , <code>choose_digits_zero</code> , <code>choose_carry_zero</code> ,<br><code>Finset.sum_range_mul</code> , <code>padicValNat.choose_lt_sq</code> | 2 |
| PadicBridge  | <code>PInt</code> , <code>PDvd</code> , <code>PCong</code> , <code>PCong.of_zmod</code>                                                                                                                    | 2 |
| Apéry        | <code>apery_lucas</code> , <code>apery_lucas_digits</code>                                                                                                                                                 | 3 |
| BrownZudilin | <code>Q_lucas</code> , <code>Q_lucas_digits</code>                                                                                                                                                         | 3 |
| Letters      | <code>Letter</code> , <code>K_descent</code> , <code>mon_descent</code>                                                                                                                                    | 4 |
| TheoremLB    | <code>theorem_LB</code>                                                                                                                                                                                    | 4 |
| Instances    | <code>bMin</code> , <code>bMin_lucas</code> , <code>bFramel</code> , <code>bFramel_lucas</code>                                                                                                            | 5 |
| MinimalForm  | <code>bApery</code> , <code>bMin_rec</code> , <code>bMin_eq_bApery</code> , <code>apery_b_harmonic_closed_form</code> ,<br><code>bApery_lucas</code>                                                       | 5 |
| BZFactor12   | <code>dlcm</code> , <code>dlcm_two_pow_five</code> , <code>BZ_P2</code> , <code>BZ_P2_not_integral</code> ,<br><code>BZ_P2_factor_twelve</code> , <code>BZ_twelve_minimal</code>                           | 6 |
| Kummer       | seven carry-counting lemmas, headed by <code>padicValNat.choose_digits</code>                                                                                                                              | — |

Four honest qualifications, stated here rather than buried.

- (a) *Nothing about the value  $\zeta(3)$  is formalised anywhere in this development.* There are no limits, no irrationality statements, and not even the convergence  $b_n/a_n \rightarrow \zeta(3)$ . Every theorem below is a congruence, an integrality statement or an identity between explicitly defined sequences of rationals. The word  $\zeta(3)$  occurs in this paper only as a label for which sequences are under discussion.
- (b) *Which sequence “ $b_n$ ” means.* In Lean,  $b_n$  is pinned by Apéry’s recurrence together with the initial values  $b_0 = 0$ ,  $b_1 = 6$  — the sequence-theoretic characterisation — and *not* by the classical double-sum expression with its inner alternating sum. That the two agree is proved on paper [11] and is *not* formalised.
- (c) *The character is never exercised.* Both instances of the general second-row theorem instantiate its character parameter at 1. The theorem carries a character, but no instance with a nontrivial character is proved, here or on paper.
- (d) *One module is unused.* The `Kummer` module is a carry-counting toolkit that no result below consumes; it is listed for completeness.

Finally, one sentence on what is *not* here. The same programme has produced further material — a master depth law supported by exact sweeps over the fifteen sporadic families, harmonic decompositions for several of them, a transfer principle relating archimedean and Volkenborn linear forms, and denominator results for the Brown–Zudilin ladder — none of which is machine-checked; it is reported in the companion papers [11, 12, 13], and nothing in this paper depends on any of it.

**Notation.**  $p$  is a prime,  $v_p$  the  $p$ -adic valuation on  $\mathbb{Q}$  with  $v_p(p) = 1$ ,  $d_n = \text{lcm}(1, \dots, n)$ , and  $H_y^{(r)} = \sum_{m=1}^y m^{-r}$ . In Lean, primality is carried by the instance `[Fact p.Prime]`, and all sums are `Finset` sums over `range`.

## 2. SETUP: THE TWO BRIDGES

Two small pieces of infrastructure carry everything else: a congruence relation on  $\mathbb{Q}$ , and a digitwise form of Lucas’ theorem.

**2.1. Congruences between rationals.** Second-row statements are congruences between *rational* numbers with controlled denominators, so the ambient relation cannot be `ZMod p` equality. We use the  $p$ -adic norm on  $\mathbb{Q}$  directly.

**Definition 2.1** (module `PadicBridge`). For  $x, y \in \mathbb{Q}$  put

$$\text{PInt } p \ x : \iff |x|_p \leq 1, \quad \text{PDvd } p \ x : \iff |x|_p < 1, \quad \text{PCong } p \ x \ y : \iff \text{PDvd } p \ (x - y),$$

where  $|\cdot|_p$  is `Mathlib`'s `padicNorm`. So `PInt` is  $p$ -integrality, i.e. membership of the local ring  $\mathbb{Z}_{(p)}$ , and `PCong`  $p \ x \ y$  is  $x \equiv y \pmod{p}$  inside  $\mathbb{Z}_{(p)}$ .

This relation is not vacuous: since  $|x - y|_p < 1$  forces  $|x - y|_p \leq 1$ , a congruence implies  $p$ -integrality of either side as soon as the other is  $p$ -integral (`PCong.pInt`). The module proves the expected closure properties — sums, products by  $p$ -integral constants, `Finset` sums — and one bridge lemma, `PCong.of_zmod`: an equality of integers in `ZMod p` entails `PCong p` of their images in  $\mathbb{Q}$ . That lemma is what lets the integer-valued first row be used inside rational-valued second-row proofs.

## 2.2. Lucas' theorem, digitwise.

**Lemma 2.2** (`Core.choose_digits`). *Let  $p$  be prime,  $a, b \geq 0$  and  $r, s < p$ . Then  $\binom{ap+r}{bp+s} \equiv \binom{a}{b} \binom{r}{s} \pmod{p}$ .*

The Lean proof specialises `Mathlib`'s `Choose.choose_modEq_choose_mod_mul_choose_div` and computes the four base- $p$  division and remainder terms explicitly. Two consequences are used repeatedly: `choose_digits_zero`, that  $\binom{n}{k} \equiv 0 \pmod{p}$  whenever the low digit of  $n$  is strictly below the low digit of  $k$ ; and `choose_carry_zero`, the carry-annihilation form — if  $p \leq x$ ,  $y < p$  and  $x - p < y$  then  $\binom{x}{y} \equiv 0 \pmod{p}$ , because the digits of  $x$  are  $(1, x - p)$  and those of  $y$  are  $(0, y)$ .

The third ingredient is a combinatorial identity, `Finset.sum_range_mul`: for any commutative monoid,  $\sum_{k < mn} f(k) = \sum_{i < m} \sum_{j < n} f(in + j)$ . This is what splits a summation range into base- $p$  blocks, and it is used in every proof in §3 and §4. Finally `padicValNat_choose_lt_sq` records that for  $n < p^2$  and  $k \leq n$  Kummer's carry count collapses to a single indicator,  $v_p \binom{n}{k} = [p \leq k \bmod p + (n - k) \bmod p]$ .

## 3. THE FIRST ROW

**Theorem 3.1** (`Apéry.apéry_lucas`, `Apéry.apéry_lucas_digits`). *Let  $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$  and  $a_n = \sum_{k=0}^n A(n, k)$ . For every prime  $p$ , every  $a \geq 0$  and every  $r < p$ ,*

$$a_{ap+r} \equiv a_a a_r \pmod{p}, \quad \text{and} \quad a_n \equiv \prod_i a_{n_i} \pmod{p}$$

for the base- $p$  digits  $(n_i)$  of  $n$ .

*The verified argument.* Four steps. (1) Extend the summation range from  $\{0 \leq k \leq ap + r\}$  to the full block  $\{0 \leq k < (a + 1)p\}$ ; this is an equality of integers, not a congruence, because  $A(n, k) = 0$  for  $k > n$ . (2) Split the block into digits by `Finset.sum_range_mul`. (3) Factor each summand modulo  $p$  by Lemma 2.2 applied to both binomials, which gives  $A(ap + r, bp + s) \equiv A(a, b)A(r, s)$ . (4) The resulting double sum is a product of two independent sums,  $\sum_b A(a, b) \cdot \sum_{s < p} A(r, s)$ ; the first factor is  $a_a$  on the nose, and the second is  $a_r$  because the terms with  $r < s < p$  carry a vanishing  $\binom{r}{s}$ . The digit-product form is a strong induction on  $n$  using `Nat.digits_def`, and it is available for *all*  $a \geq 0$ , which is what makes the induction go through.  $\square$

Note the absence of hypotheses: no  $p \geq 5$ , no  $a < p$ , no  $p$ -unit condition. The same argument, with a two-index summand, gives the analogue for the double sum that carries the Brown–Zudilin cellular approximations to  $\zeta(5)$  [2].

**Theorem 3.2** (`BrownZudilin.Q_lucas`, `BrownZudilin.Q_lucas_digits`). *Let*

$$T(n, k, l) = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}, \quad Q_n = \sum_{k,l=0}^n T(n, k, l).$$

*For every prime  $p$ , every  $a \geq 0$  and every  $r < p$ , one has  $Q_{ap+r} \equiv Q_a Q_r \pmod{p}$ , and  $Q_n \equiv \prod_i Q_{n_i} \pmod{p}$  over base- $p$  digits.*

*The verified argument.* As for Theorem 3.1, with both ranges extended and both split; the summand factorises as  $T(ap+r, bp+s, cp+t) \equiv T(a, b, c)T(r, s, t) \pmod{p}$ , the three wide binomials  $\binom{n+k}{n}$ ,  $\binom{n+l}{n}$ ,  $\binom{n+k+l}{n}$  being handled in their carrying regimes by `choose_carry_zero` and `choose_digits_zero`. One extra step is needed relative to the single-index case: after factorisation the four sums appear in the order  $b, s, c, t$ , and a `Finset.sum_comm` is required to bring the high-digit pair  $(b, c)$  and the low-digit pair  $(s, t)$  together before the quadruple sum separates into a product.  $\square$

#### 4. THE SECOND ROW: HARMONIC LETTERS AND THE ABSTRACT THEOREM

##### 4.1. Letters.

**Definition 4.1** (module `Letters`). For a completely multiplicative  $\chi : \mathbb{N} \rightarrow \mathbb{Z}$  and  $r, y \in \mathbb{N}$  put

$$K_\chi^{(r)}(y) = \sum_{m=0}^y \frac{\chi(m)}{m^r} \in \mathbb{Q}.$$

The sum is over `range (y+1)`; the term  $m = 0$  is  $\chi(0)/0^r = 0$  in  $\mathbb{Q}$  for  $r \geq 1$ , so the object is the usual truncated twisted zeta value  $\sum_{m=1}^y \chi(m)m^{-r}$ , and no `Finset.Icc` reindexing is ever required. A *letter* is a structure bundling  $\chi$  together with a proof of complete multiplicativity, a positive degree  $r$ , and an argument form  $x : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$ ; a *monomial* is a `List Letter`, with `monDeg` the sum of the degrees and `monChi`  $M p = \prod_{\ell \in M} \chi_\ell(p)$ .

We deliberately do not use Mathlib’s `DirichletCharacter`, which would force a modulus and a `ZMod`-valued target; complete multiplicativity as a hypothesis field is exactly what the descent needs, and it admits  $\chi \equiv 1$ .

**Lemma 4.2** (`Letters.K_descent`). *Let  $\chi$  be completely multiplicative,  $r \geq 1$  and  $y \geq 0$ . Then*

$$p^r K_\chi^{(r)}(y) \equiv \chi(p) K_\chi^{(r)}(\lfloor y/p \rfloor) \pmod{p}.$$

*The verified argument.* Split the sum at  $p \mid m$ . The  $p$ -divisible layer reproduces the right-hand side *exactly*, before any congruence is taken: reindexing  $m = jp$  and using complete multiplicativity,  $\sum_{j \leq \lfloor y/p \rfloor} \chi(jp)(jp)^{-r} = \chi(p)p^{-r} K_\chi^{(r)}(\lfloor y/p \rfloor)$ . The remaining layer has  $p$ -integral denominators, so after multiplication by  $p^r$  with  $r \geq 1$  it is  $p$ -divisible. Hence the difference of the two sides is  $p^r$  times a  $p$ -integral number.  $\square$

Lemma 4.2 is where the character enters, and it is the only place. Multiplying the lemma over the letters of a monomial gives `mon_descent`: if every argument satisfies  $\lfloor x(n, k)/p \rfloor < p$ , then

$$p^{\text{monDeg } M} \cdot M(n, k) \equiv \text{monChi } M p \cdot M^\downarrow(n, k) \pmod{p},$$

where  $M^\downarrow$  is  $M$  with every argument  $x$  replaced by  $\lfloor x/p \rfloor$ . The Lean proof is an induction over the list, using that each factor is  $p$ -integral so that the cross terms in the product of two congruences are absorbed.

**4.2. The abstract second-row theorem.** Fix a summand  $S : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{Z}$ , a finite index set  $J$ , coefficients  $c : J \rightarrow \mathbb{Q}$  and monomials  $\text{mon} : J \rightarrow \text{List Letter}$ , and put

$$w(n, k) = \sum_{j \in J} c_j \text{mon}_j(n, k), \quad A(n) = \sum_{k \leq n} S(n, k), \quad B(n) = \sum_{k \leq n} S(n, k) w(n, k),$$

the last two being **Arow** and **Brow** in the development. The theorem below carries a character value  $\chi_p$  as a parameter, and since both instances proved here set it to 1 it is worth saying why it is there: in the companion work [11] the untwisted form of this congruence is found to fail for seven of the fifteen sporadic Apéry-like families, at precisely the primes where a quadratic character takes the value  $-1$ , and Lemma 4.2 — formalised here as **K\_descent** — is the mechanism that produces the compensating factor. We state that as motivation for the shape of the theorem, and claim none of it here.

**Theorem 4.3** (**TheoremLB.theorem\_LB**). *Let  $p$  be prime and let  $w \in \mathbb{N}$ ,  $\chi_p \in \mathbb{Z}$ . Assume*

- (H1):  $S(ap + r, bp + s) \equiv S(a, b)S(r, s) \pmod{p}$  for all  $a, b \geq 0$  and all  $r, s < p$ ;
- (H2):  $S(n, k) = 0$  whenever  $n < k$ ;
- (H3): for all  $r, s < p$  and  $b \leq a$  with  $S(r, s) \not\equiv 0 \pmod{p}$ , every argument form  $x$  occurring in the weight satisfies  $\lfloor x(ap + r, bp + s)/p \rfloor = x(a, b)$ ;
- (H4): every argument form satisfies  $x(n, k) \leq n$  whenever  $k \leq n$ ;
- (H4c): every coefficient  $c_j$  is  $p$ -integral;
- (Hw): every monomial has total degree exactly  $w$ ;
- (H5):  $\text{monChi mon}_j p = \chi_p$  for every  $j \in J$ .

Then for all  $a < p$  and  $r < p$ ,

$$p^w B(ap + r) \equiv \chi_p B(a) A(r) \pmod{p}.$$

*The verified argument.* Write  $n = ap + r$ , so  $n < p^2$  and  $\lfloor n/p \rfloor = a$ . By (H4) every argument of the weight at  $(n, k)$  is  $\leq n < p^2$ , so its descent  $\lfloor x/p \rfloor$  is  $< p$ ; with (H4c) this makes  $p^w w(n, k)$   $p$ -integral for every  $k$ , and  $w(a, b)$   $p$ -integral as well. Extend the range of  $B(n)$  to the full block  $\{0 \leq k < (a + 1)p\}$  — an equality, by (H2) — and split it into digits. On the layer where  $S(r, s) \equiv 0 \pmod{p}$ , (H1) makes each term  $\equiv 0$ , because it is  $p$  times a  $p$ -integral number. On the complementary layer, (H3) identifies the descended weight with  $w(a, b)$ , **mon\_descent** and (Hw), (H5) give  $p^w w(n, k) \equiv \chi_p w(a, b)$ , and the double sum factors as  $(\sum_{b \leq a} S(a, b) w(a, b)) (\sum_{s < p} S(r, s)) = B(a) A(r)$ , the second factor again by (H2).  $\square$

*Remark 4.4* (what the formal hypotheses cost, and save). Two features of the hypothesis list are worth flagging for anyone instantiating it. (H1) is an *unconditional* congruence, required for all  $a, b$  rather than only on a surviving set; in the carrying regime both sides vanish modulo  $p$ , so the usual case dichotomy is absorbed, and in consequence no product-region argument occurs anywhere in the formal proof: the double sum factors because a full block splits, and the extension to the full block is an equality of integers. (H3) is conditioned on  $S(r, s) \not\equiv 0 \pmod{p}$ , which is exactly what excludes a borrow in an argument such as  $n - k$ : an instance *derives* the no-borrow condition instead of describing the surviving set. (H5) asks that the product of the letters' characters at  $p$  be the same for every monomial, which is weaker than requiring a fixed number of twisted letters per monomial and permits mixed conductors.

## 5. TWO INSTANCES, AND A CLOSED FORM

### 5.1. The weight-three instance.

**Definition 5.1** (**Instances.bMin**).  $B_{\min}(n) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)})$ .

**Theorem 5.2** (`Instances.bMin_lucas`). *For every prime  $p$  and all  $a, r < p$ ,*

$$p^3 B_{\min}(ap + r) \equiv B_{\min}(a) a_r \pmod{p}.$$

*The verified argument.* Theorem 4.3 with  $S(n, k) = A(n, k)$ ,  $J$  of size two, coefficients  $(2, -1)$ , monomials  $[H^{(3)}]$  at the arguments  $x(n, k) = n$  and  $x(n, k) = k$ , degree  $w = 3$  and  $\chi_p = 1$ . (H1) is the digit factorisation of  $A$  established for Theorem 3.1; (H2) is  $\binom{n}{k} = 0$  for  $k > n$ ; (H3) holds because  $\lfloor (ap + r)/p \rfloor = a$  and  $\lfloor (bp + s)/p \rfloor = b$ ; (H4) because both arguments are  $\leq n$ ; (H4c) because the coefficients are the integers  $2, -1$ .  $\square$

There is no hypothesis  $p \geq 5$  and no hypothesis  $a \geq 1$  — both of which the informal analysis of this instance had carried. The first is removable precisely because the coefficients of the minimal weight are integers, so (H4c) is vacuous; the second because  $B_{\min}(0) = 0$  and  $p^3 B_{\min}(r)$  is  $p$ -divisible for  $r < p$ .

## 5.2. The weight-two instance.

**Theorem 5.3** (`Instances.bFranel_lucas`). *Let  $f_n = \sum_k \binom{n}{k}^3$  be the Franel numbers (OEIS A000172) and*

$$B_{\mathbf{F}}(n) = \sum_{k=0}^n \binom{n}{k}^3 \left( \frac{1}{4} H_k^{(2)} + \frac{3}{4} H_k^2 - \frac{3}{4} H_k H_{n-k} \right).$$

*For every prime  $p \neq 2$  and all  $a, r < p$ , one has  $p^2 B_{\mathbf{F}}(ap + r) \equiv B_{\mathbf{F}}(a) f_r \pmod{p}$ .*

*The verified argument.* Theorem 4.3 with  $S(n, k) = \binom{n}{k}^3$ , three monomials —  $H_k^{(2)}$ ,  $H_k H_k$  and  $H_k H_{n-k}$  — coefficients  $\frac{1}{4}, \frac{3}{4}, -\frac{3}{4}$ , degree  $w = 2$  and  $\chi_p = 1$ . The only hypothesis needing care is (H3) for the argument  $n - k$ : it is here that the side condition  $S(r, s) \not\equiv 0 \pmod{p}$  does its work, giving  $\binom{r}{s} \neq 0$ , hence  $s \leq r$ , hence  $\lfloor (n - k)/p \rfloor = a - b$  with no borrow. The hypothesis  $p \neq 2$  enters only through (H4c), the coefficient denominators being 4.  $\square$

**5.3. The closed form.** Theorem 5.2 is a statement about the explicit sum  $B_{\min}$ . The following identifies that sum with Apéry's companion sequence.

**Definition 5.4** (`MinimalForm.bApery`).  $b : \mathbb{N} \rightarrow \mathbb{Q}$  is defined by  $b_0 = 0$ ,  $b_1 = 6$  and  $(n + 1)^3 b_{n+1} = P(n) b_n - n^3 b_{n-1}$  with  $P(n) = 34n^3 + 51n^2 + 27n + 5$ ; the leading coefficient never vanishes, so this determines  $b_n$  uniquely.

This is the sequence-theoretic characterisation of Apéry's companion: the recurrence together with two initial values. It is *not* the classical double-sum expression, in which the weight attached to  $\binom{n}{k}^2 \binom{n+k}{k}^2$  is  $H_n^{(3)} + \sum_{m \leq k} (-1)^{m-1} (2m^3 \binom{n}{m} \binom{n+m}{m})^{-1}$ ; that the two agree is proved on paper in [11] and is not formalised. Every occurrence of  $b_n$  below means Definition 5.4.

**Theorem 5.5** (`MinimalForm.bMin_eq_bApery` and `apery_b_harmonic_closed_form`). *For every  $n \geq 0$ ,*

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}) = b_n.$$

*The verified argument.* Both sides satisfy the same order-two recurrence and agree at  $n = 0, 1$ ; since the leading coefficient  $(n + 1)^3$  never vanishes, a two-step induction concludes. That  $B_{\min}$  satisfies the recurrence (`bMin_rec`) is the substance. Everything is driven by four multiplicative absorption identities (10a–10d) which, writing  $n = m + 1$  and  $\Phi(m, k) = \binom{m+2}{k}^2 \binom{m+k}{k}^2 / ((m + 1)^2 (m + 2)^2)$ , express  $A(m, k)$ ,  $A(m + 1, k)$ ,  $A(m + 2, k)$  and  $\Phi(m, k + 1) (k + 1)^4$  as  $\Phi(m, k)$  times explicit quadratic factors. These hold for *all*  $k \geq 0$ , including the degenerate range

$k > n$ , so no 0/0 and no boundary case ever arises, and after them every step is  $\Phi(m, k)$  times a polynomial identity in  $\mathbb{Q}[m, k]$  discharged by `ring`. Two certificates are then exhibited: a Zeilberger certificate  $G = \Phi k^4 T$  with  $T(n, k) = 4(2n+1)(2k^2 - 3k - 4n^2 - 4n)$ , and a Gosper certificate  $K = \Phi k^4 U / (n^2(n+1)^2)$  with  $U(n, k) = -4(k-1)(2n+1)(2n^2 + 2n + 1 - k)$  which absorbs the weight correction. They combine into a *single* telescope: with  $Gd = \Phi k T$  and

$$\Psi(m, k) = (2H_{m+1}^{(3)} - H_k^{(3)})G(m, k) + K(m, k) + Gd(m, k),$$

the  $k$ -summand of  $(Lb)(m+1)$  is exactly  $\Psi(m, k+1) - \Psi(m, k)$  (`star`), so `Finset.sum_range_sub` collapses the whole sum to  $\Psi(m, m+3) - \Psi(m, 0) = 0$ . There is no Abel summation and no boundary bookkeeping beyond the two evaluations  $\Psi(m, 0) = \Psi(m, m+3) = 0$ .  $\square$

*Remark 5.6* (what Theorem 5.5 does and does not say). What is verified is an identity between an explicit binomial-harmonic sum and the solution of an explicit recurrence. That  $b_n/a_n \rightarrow \zeta(3)$ , that  $b_n$  coincides with Apéry's classical nested-sum expression, and that  $d_n^3 b_n \in \mathbb{Z}$  are *not* formalised and are not asserted here.

Combining the last two theorems removes the explicit sum from the statement altogether, and gives the congruence in the form a reader of §1 would want it: a statement about the solution of Apéry's recurrence.

**Theorem 5.7** (`MinimalForm.bApéry_lucas`). *Let  $b$  be as in Definition 5.4 and  $a_n$  as in Theorem 3.1. For every prime  $p$  and all  $a, r < p$ ,*

$$p^3 b_{ap+r} \equiv b_a a_r \pmod{p}.$$

*The verified argument.* Theorem 5.2 transported along Theorem 5.5: rewrite both occurrences of  $b$  as  $B_{\min}$  and apply `bMin_lucas`.  $\square$

*Remark 5.8* (status in the literature). The recurrence of Definition 5.4 is Apéry's, and that his companion sequence satisfies it is classical; a complete creative-telescoping proof is given by Schneider [8], and the identity goes back to the reorganisation of Apéry's proof reported by van der Poorten [10, §8]. What we have not been able to locate in the literature is the weight  $2H_n^{(3)} - H_k^{(3)}$  of Theorem 5.5 — in particular it is not among the closed forms recorded for OEIS A059415. The nearest published relative we found is Nesterenko's Lemma 1 [7], which amounts to the longer weight  $H_k^{(3)} + (2H_k - H_{n+k} - H_{n-k})H_k^{(2)}$ .

## 6. AN ARITHMETIC NOTE ON THE BROWN–ZUDILIN DENOMINATORS

Brown and Zudilin [2] construct cellular approximations to  $\zeta(5)$  and record, as an experimental observation, the inclusions  $Q_n, d_n^2 d_{2n} \hat{P}_n, d_n^5 P_n \in \mathbb{Z}$  for the totally symmetric member of their family. They also print the value  $P_2 = 1190161/384$ .

The following is an arithmetic consequence of that printed numeral and nothing else. We emphasise what is and is not formalised: the module `BZFactor12` takes  $P_2 = 1190161/384$  as a *quoted rational literal*, a hypothesis of the file; no cellular integral, no period, and no part of the construction that produced the number is formalised or verified. If the printed value were a typographical error, the statements below would say nothing about the corrected value.

**Theorem 6.1** (`BZ_P2_not_integral`, `BZ_P2_factor_twelve`, `BZ_twelve_minimal`). *Let  $d_n = \text{lcm}(1, \dots, n)$  be computed from `Nat.lcm` and let  $P_2 = 1190161/384$ . Then  $d_2^5 = 32$  and*

$$d_2^5 P_2 = \frac{1190161}{12} \notin \mathbb{Z}, \quad 12 d_2^5 P_2 = 1190161 \in \mathbb{Z},$$

*and moreover 12 is minimal: every  $c \in \mathbb{N}$  with  $c d_2^5 P_2 \in \mathbb{Z}$  is divisible by 12.*

*The verified argument.*  $d_2 = \text{lcm}(1, 2) = 2$  and  $d_2^5 = 32$  are computed by `decide` from the definition of  $d_n$  by recursion on `Nat.lcm`, not hard-coded;  $32 \cdot 1190161/384 = 1190161/12$  by `norm_num`. If  $1190161/12$  were an integer  $z$  then  $1190161 = 12z$  in  $\mathbb{Z}$ , which `omega` refutes. For minimality,  $c \cdot 1190161/12 = z$  gives  $1190161c = 12z$  in  $\mathbb{Z}$ , and since  $1190161 \equiv 1 \pmod{12}$  one gets  $12 \mid c$ ; the Lean proof exhibits the witness  $c \cdot (-99180) + z$  directly.  $\square$

So with the printed  $P_2$ , the inclusion  $d_n^5 P_n \in \mathbb{Z}$  fails at  $n = 2$ , and fails by exactly  $12 = 2^2 \cdot 3$ : no smaller multiplier repairs it, and 12 does. We record it in this form because it is exactly the part of the matter that a kernel can settle. Whether 12 is the right global constant for all  $n$ , and where its two prime factors come from, is a computational and structural question treated in [12]; nothing about it is claimed here.

## 7. BUILDING AND CHECKING THE DEVELOPMENT

The development is 2107 lines across the ten `ZetaLucas/` modules (2117 including the root import file), and depends only on Mathlib [5, 6]. The toolchain is pinned in `lean-toolchain` and the Mathlib revision in `lake-manifest.json`:

```
leanprover/lean4:v4.33.0-rc1
mathlib rev cd580e54f1a6b46063824e80cec92f64692cbe78

lake exe cache get
lake build
```

With Mathlib prebuilt this replays in seconds and reports `Build completed successfully`. Three independent checks establish the claim made in §1, and a reader is invited to run all three:

- (1) *No holes.* `grep -rn "sorry\|admit\|native_decide" ZetaLucas/` returns only one hit, inside a prose comment in `Kummer.lean` (“two digits admit at most one carry”). No declaration in the development is stated with `sorry`, and no proof uses `native_decide`, so no statement rests on compiled-code evaluation.
- (2) *No extra axioms.* `#print axioms` on each declaration named in this paper returns `[propext, Classical.choice, Quot.sound]` and nothing else — in particular no `sorryAx`.
- (3) *The statements are the ones printed here.* Each theorem above cites its declaration; the Lean statements are short and were written to be read.

Two small points of hygiene. The development contains a handful of `#eval` normalisation checks (Apéry and Franel values, the first terms of  $B_{\min}$  and  $B_{\mathbf{F}}$ , and digitwise sweeps of the two instances at  $p = 5, 7$ ). These are *evaluations*, not *kernel proofs*, they are marked as such in the source, and no statement in this paper depends on them. Separately, five anonymous `examples` are kernel-checked: in `Apery.lean`, by `decide`, that  $a_0, \dots, a_5 = 1, 5, 73, 1445, 33001, 819005$  and that  $a_{5a+r} \equiv a_a a_r \pmod{5}$  for all  $a, r < 5$ ; and in `MinimalForm.lean`, by `norm_num`, that  $b_2 = 351/4$  and  $b_3 = 62531/36$  from Definition 5.4 and that  $B_{\min}(2) = 351/4$  from the sum.

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- Companion papers of the same programme. None of their contents is machine-checked, and nothing in the present paper depends on them:*
- [11]  *$\chi$ -twisted Lucas congruences for the second solutions of Apéry-like recurrences* (companion paper).
  - [12] *Sharp denominators for the Brown–Zudilin cellular approximations to  $\zeta(5)$*  (companion paper).
  - [13] *A transfer principle for  $p$ -adic hypergeometric linear forms, and a margin map* (companion paper).